Thambawita Maddumage Sajith Priyankara

Email: tpriyank@ttu.edu — tmspriyankara@gmail.com
Website: https://tmspriyankara.github.io — Google Scholar: Profile
Mobile: (806) 559–4870

Summary

Ph.D. Candidate in Statistics, Texas Tech University, USA (September 2021 – July 2026).
M.S. in Statistics, Texas Tech University, USA (September 2021 – December 2023).
B.Sc. in Statistics, University of Peradeniya, Sri Lanka (November 2014 – February 2019).
Research & Teaching Experience – Over 8 years of research experience in statistical methodology for biological and health-related data, including computational biology and epidemiologic applications, with advanced data analysis across multiple projects. Over 3 years of strong teaching experience across diverse undergraduate courses.

Research Projects

- **Ph.D. Dissertation:** Ligand-protein binding prediction using geometric-aware, object-oriented data representations and high-dimensional modeling techniques.
  Skills: *R, functional data analysis, compositional data analysis, statistics on manifolds, multivariate modeling, machine learning*
  - Designed novel structural and compositional representations of protein binding sites using an object-oriented functional data framework.
  - Applied constrained and unconstrained functional data analysis, statistics on manifolds, and multivariate statistical modeling.
  - Developed similarity and distance measures between high-dimensional covariance objects for downstream classification tasks.
  - Developed predictive models for ligand–protein binding using machine learning methods, including random forests, LASSO-penalized multinomial logistic regression, and Mahalanobis-distance–based nearest mean classifiers.
  - Two first-author manuscripts from this work are currently in final draft preparation.
- **Worked on a project to model air pollution in Kandy, Sri Lanka, using smart sensors**.
  - The project was funded by the NHMRC Centre for Air Pollution, Energy and Health Research (CAR), and a collaboration with Queensland University of Technology.
- **Air Pollution and Hospitalization: Published first-author paper** analyzing short-term effects of PM2.5/PM10 on respiratory disease hospitalizations in Kandy, Sri Lanka.
  Skills: *R, time series analysis, Generalized Additive Models (GAMs), epidemiology, environmental health*
  - Applied GAMs with quasi-Poisson to model daily respiratory hospitalizations as a function of PM2.5/PM10, controlling for long-term trends using spline smoothing.
  - Evaluated lagged exposure effects (lag0–lag5) and conducted age and sex stratified analyses.
  - Performed high vs low pollution period comparisons using rate ratios to quantify seasonal risk differences.
- **BMI Trajectories and Asthma Risk: Published second-author paper** based on longitudinal analysis of parental BMI trajectories and asthma risk in offspring using Tasmanian Longitudinal Health Study data.
  Skills: *STATA, R, longitudinal data analysis, epidemiology, biostatistics*

  - This study aimed to investigate the association between preconception BMI trajectories until adolescence in parents and risk of asthma in their offspring.
  - Used group-based trajectory modeling to analyze height and weight data from parents at various time points, obtained from school medical records.
  - Multinomial logistic regression models adjusted for potential confounders were used to estimate the association between parents' BMI trajectories and risk of asthma phenotypes. A mediation analysis and a sensitivity analysis were also conducted
- **National Burden of Disease Study (Sri Lanka):** Estimating Years of Life Lost (YLL) using national civil registration data. (First author manuscript is in final draft stage)
  Skills: *R, public health data analysis*
- **Undergraduate final year research project**: Early recognition of premature birth: A comparative analysis.
  - The main purpose of this study was to identify a statistical method to early predict preterm birth with the use of uterine electromyography (EMG) signals of pregnancies.
  - Extracted features from EMG signals and addressed class imbalance using SMOTE. Built and evaluated classification models, including logistic regression, SVM, and decision trees with PCA-based dimensionality reduction.

  Skills: *R, machine learning.*

Professional and Teaching Experience

- **Graduate Part-Time Instructor (Instructor of Record)**, Department of Mathematics and Statistics, Texas Tech University, Aug. 2022 – Present
  Responsible for full course delivery, exam preparation, grading, and student support for 80-120 undergraduate students per section. Courses taught:
  - MATH 2300 – Statistical Methods
  - MATH 1330 – Introductory Mathematical Analysis I
  - MATH 1331 – Introductory Mathematical Analysis II
- **Graduate Teaching Assistant**, Texas Tech University, Aug. 2021 – Aug. 2022
- **Research Assistant**, National Institute of Fundamental Studies, Sri Lanka, 2019 – Aug. 2021

Publications

Journal Articles

1. Priyankara, S., et al. (2021). Ambient PM2.5 and PM10 Exposure and Respiratory Disease Hospitalization in Kandy, Sri Lanka. *Int. J. Environ. Res. Public Health.* (First author)

2. Bowatte, G., Bui, D.S., Priyankara, S., et al. (2022). Parental preconception BMI trajectories and asthma in offspring. *J. Allergy Clin. Immunol.* (Second author; equal contribution by first two authors)

3. Senarathna, M., Priyankara, S., et al. (2022). Measuring Traffic-Related Air Pollution Using Smart Sensors in Sri Lanka. *Geography, Environment, Sustainability.* (Second author)

Selected Abstracts and Conference Presentations

1. Priyankara, S., Ellingson, L. (2024). Analysis of Covariance Matrix Distances in Protein Binding Sites. *Conference of Texas Statisticians.* (First author)

2. Priyankara, S., et al. (2021). PM2.5 Exposure and Respiratory Disease Hospitalization in Sri Lanka. *Lung Science Conference – European Respiratory Society.* (First author)

3. Priyankara, S., Abeysundara, S.P. (2020). Early Recognition of Premature Birth. *International Conf. on Environmental and Medical Statistics*. (First author)
4. Bowatte, G., Bui, D., Priyankara, S., et al. (2020). Parental BMI Trajectories and Offspring Asthma Risk. *TSANZ Annual Scientific Meeting*. (Third author)
5. Yasaratne, D., Priyankara, S., et al. (2021). Asthma in Pregnancy: A Case-Control Study. *European Respiratory Journal*. (Co-author)

Programming & Data Analysis

- R, Python, STATA, MATLAB, C

Professional Service and Leadership

- Member of the American Statistical Association (ASA)
- Volunteer, Refuge Services, Lubbock, Texas (June–July 2023)
- President of Statistical circle, University of Peradeniya (2017-2018)